

EGR 345 HW 1: Solving Differential Equations using Laplace

Use the Laplace transform to solve the following differential equations:

1. $\dot{y} + 7y(t) = 0$, $y(0) = 3$
2. $3\dot{y} + 5y(t) = 0$, $y(0) = 2$
3. $\dot{y} + 4y(t) = 1(t)$, where $1(t)$ is a unit step input that switches “on” at $t = 0$:

$$1(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

4. $\ddot{y} + 2\dot{y} + 49y(t) = 0$, $y(0) = 4$ and $\dot{y}(0) = 0$
5. $\ddot{y} + 2\dot{y} + 49y(t) = 0$, $y(0) = 0$ and $\dot{y}(0) = 3$